



Saeed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 3

Obtained Marks: _____

Introduction to Data Science

Assignment # 04

Continuation of Assignment # 03

AQI-Based Adaptive Traffic-Control System Using Reinforcement Learning

Important Continuation Note: Assignment 4 must be added as the next phase of the same AQI project completed in Assignment 3. Students should reuse their cleaned dataset, AQI categories, visualizations, and GitHub repository from Assignment 3 instead of starting a completely new unrelated project.

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1. Connection with Assignment 3

This assignment is a direct continuation of Assignment 3, where the Global Urban Air Quality Index Dataset (2015-2025) was analyzed through data cleaning, visualization, AQI category creation, KNN classification, Naive Bayes, K-Means clustering, and PCA. In Assignment 4, the same cleaned dataset and AQI categories are reused to build a Reinforcement Learning agent that learns adaptive traffic control decisions based on AQI conditions.

Outputs reused from Assignment 3:

Assignment 3 Output	How it is Used in Assignment 4
Cleaned AQI Dataset	Environment input for training and testing the RL agent
AQI Value column	Used to define RL states (Low / Medium / High)
City / Country / Date	Optional analysis of where and when actions are selected
Pollutant features	State context and interpretation

2. Problem Statement

This project extends the AQI data science analysis by designing a simplified reinforcement learning system for adaptive traffic control. Urban air pollution is a growing public health concern and cities need smart systems to respond to changing AQI conditions. The RL agent observes AQI conditions from the dataset and learns whether the city should apply No Restriction, Partial Restriction, or a High-Pollution Alert. The purpose is to understand how an agent can learn a decision policy using rewards based on public-health protection and traffic-disruption cost.

3. Reinforcement Learning Components

RL Component	Definition in This Assignment
Agent	The adaptive traffic-control decision maker
Environment	AQI conditions taken from the cleaned AQI dataset
State	AQI condition: Low AQI, Medium AQI, or High AQI
Action	No Restriction, Partial Restriction, or High-Pollution Alert
Reward	Numerical score encouraging suitable action and penalizing unsuitable action
Policy	Learned mapping from AQI state to the best traffic action
Episode	One training cycle where the agent interacts with 50 AQI records

4. AQI State Mapping

The 3-state design was used as recommended for an introductory course. Each state maps to a range of AQI values and represents a level of air quality risk.

State ID	State Name	AQI Range	AQI Condition
0	Low AQI	0 – 100	Good / Moderate
1	Medium AQI	101 – 200	Unhealthy for Sensitive Groups / Unhealthy
2	High AQI	201+	Very Unhealthy / Hazardous

5. Reward System

The reward table was designed to balance public-health protection and traffic disruption. A positive reward is given when the correct action is taken. Strong penalties are applied when dangerous situations (High AQI) are ignored, and smaller penalties for over-restricting traffic during clean air conditions.

AQI State	No Restriction	Partial Restriction	High-Pollution Alert	Best Action
Low AQI	+10	-2	-5	No Restriction
Medium AQI	-6	+10	+2	Partial Restriction
High AQI	-10	-3	+10	High-Pollution Alert

1. Q-Learning Implementation Summary

A simple tabular Q-learning algorithm was implemented using NumPy, Pandas, and Matplotlib. No advanced RL libraries were used. The agent was trained for 1000 episodes with 50 AQI records sampled per episode. The epsilon-greedy strategy was used for action selection, where the agent explores randomly 20% of the time and exploits its best known action 80% of the time.

Q-Learning Update Rule:

$$Q(\text{state}, \text{action}) = Q(\text{state}, \text{action}) + \alpha \times [\text{reward} + \gamma \times \max(Q(\text{next_state})) - Q(\text{state}, \text{action})]$$

Parameter	Value	Description
Alpha (α)	0.1	Learning rate — how fast the agent updates Q-values
Gamma (γ)	0.9	Discount factor — importance of future rewards
Epsilon (ϵ)	0.2	Exploration rate — 20% random, 80% best known action
Episodes	1000	Total training cycles
Steps per episode	50	AQI records sampled per episode

Training and Evaluation Results

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After 1000 training episodes the agent successfully converged to the correct policy for all three AQI states. The average reward over the last 100 episodes was 408.73, which indicates the agent is consistently

selecting the correct action.

Metric	Value
Number of states	3
Number of actions	3
Episodes used	1000
Learning rate (alpha)	0.1
Discount factor (gamma)	0.9
Exploration rate (epsilon)	0.2
Average reward (last 100 eps)	408.73

Visualizations

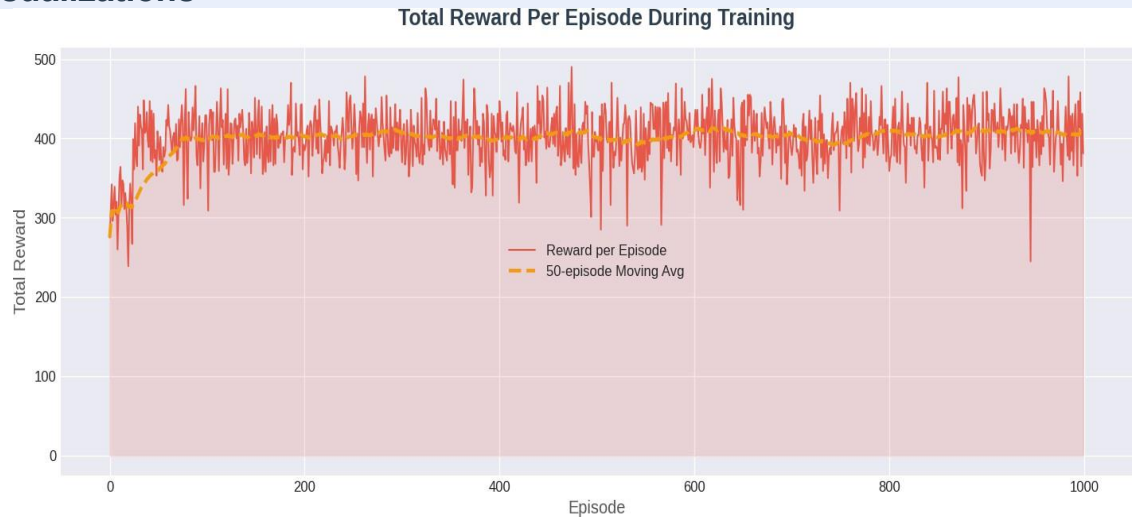


Figure 1: Total Reward Per Episode

This chart shows how the total reward changes over 1000 training episodes. The red line shows the raw reward per episode while the yellow dashed line shows the 50-episode moving average. The reward generally increases and stabilizes over time, showing the agent is learning the correct policy. Early episodes have high variation due to random exploration.

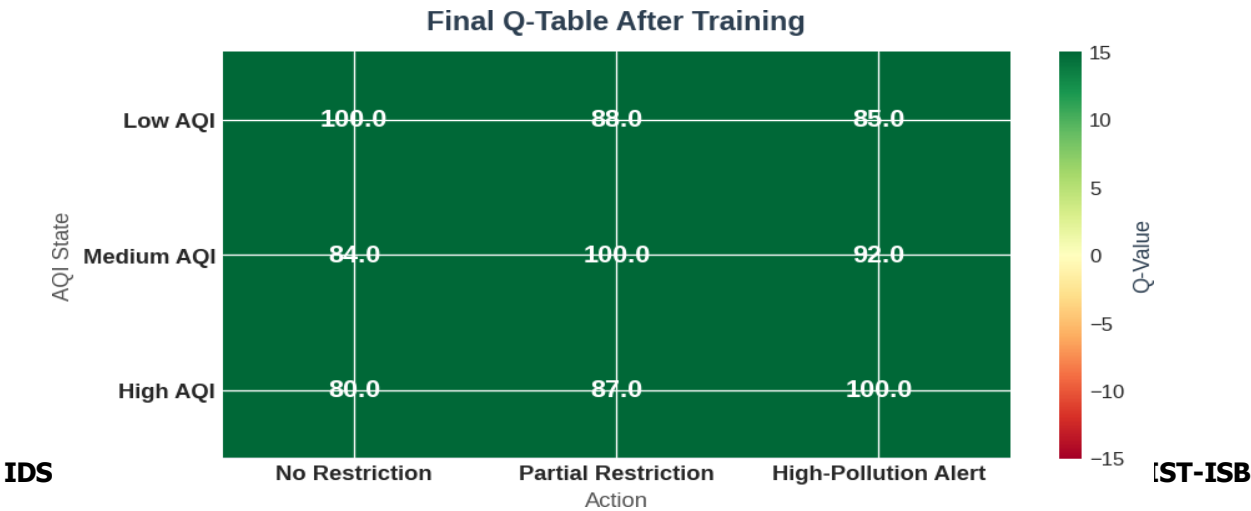


Figure 2: Final Q-Table Heatmap

This heatmap shows the Q-values for each state-action pair after training. Green cells represent high

positive values meaning the agent strongly prefers that action for that state. Red cells represent negative values meaning the agent learned to avoid those actions. The diagonal pattern confirms the agent learned the correct policy for all three states.

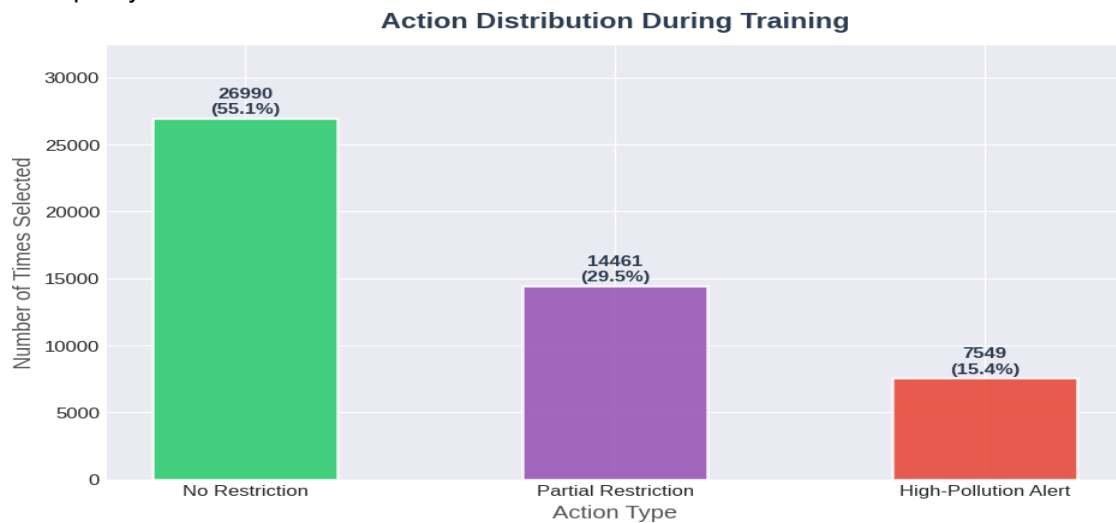


Figure 3: Action Distribution During Training

This bar chart shows how many times each action was selected across all training episodes. No Restriction was selected most often because the dataset has more Low AQI records. High-Pollution Alert was selected least often reflecting the rarer occurrence of hazardous AQI levels. The percentages confirm the distribution matches the dataset AQI profile.

GitHub Link: <https://github.com/saeedmughira-svg/Intoduction-to-data-science-Project>

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